Huawei Access Node Optical Transport Failure Recovery Procedure for BAS-ACC-73 at Basavanagudi POP

Node: BAS-ACC-73 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_1939

Description

The failure domain of Optical Transport Failure, as denoted by alarm code ALARM_1939 with critical severity on the Huawei access node BAS-ACC-73 situated at Basavanagudi POP, typically arises due to intrinsic or extrinsic factors affecting the optical signal transmission. The intrinsic factors may encompass SFP/XFP degradation, which could be attributed to the gradual deterioration of the optical components over time, potentially exacerbated by environmental conditions such as temperature fluctuations or physical stress. Extrinsic factors, on the other hand, may include fiber cuts or accidental disconnections, which disrupt the optical path, and Rx/Tx power loss, indicating a mismatch or failure in the power levels required for signal transmission. Given the critical nature of this failure, which impacts the access role of the node, it is imperative to adhere to a structured recovery procedure that aligns with Huawei's CLI dialect to ensure swift and effective resolution. The recovery process involves a series of diagnostic commands to isolate the root cause, followed by targeted resolution steps, and culminates in verification commands to confirm the restoration of optical transport functionality.

Pre-requisite Checks

- Dispatch remote hands to Basavanagudi POP for on-site inspection and intervention.

- Verify the status of redundant links to ensure no single point of failure exists.

- Check environmental conditions such as temperature and humidity within the POP to ensure they are within the operational specifications of the equipment.

Potential Root Causes

Cause	Probability
SFP/XFP Degradation	High
Fiber Cuts or Accidental Disconnections	Medium
Rx/Tx Power Loss	Low

Diagnostic Commands & Expected Outputs

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Command: display interface brief, Expected Output: List of interfaces with their current status, Description: This command provides a brief overview of all interfaces on the device, helping to identify any interfaces that are down or have errors, which could indicate the source of the optical transport failure.
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Command: display optical-attribute, Expected Output: Optical attributes of the interfaces, Description: This command displays the optical attributes such as the transmit and receive power levels, which can help in diagnosing issues related to power loss or signal degradation.

Command: display alarm all, Expected Output: List of all current alarms on the device, Description: This command lists all current alarms on the device, which can provide additional context or clues about the nature of the failure beyond the initial ALARM_1939.
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Resolution Steps

Step 1: Step: Replace Degraded SFP/XFP, Command: replace sfp slot , Description: If the diagnostic commands indicate SFP/XFP degradation, replacing the affected module with a new one can resolve the issue. The slot number should be replaced with the actual slot number of the degraded SFP/XFP as identified during diagnostics.

Step 2: Step: Repair or Replace Fiber Cuts, Command: None (Physical Intervention Required), Description: In cases where fiber cuts or disconnections are identified, physical repair or replacement of the affected fiber segment is necessary. This step requires on-site intervention and does not involve CLI commands.

Step 3: Step: Adjust Rx/Tx Power Levels, Command: set optical-power , Description: If the issue is related to Rx/Tx power loss, adjusting the power levels to the recommended settings can resolve the problem. The interface and power level should be specified according to the device's documentation and the results of the diagnostic commands.

Post-Check Verification

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Command: display interface brief, Expected Output: Interfaces should be up and without errors, Description: After implementing the resolution steps, re-running the interface brief command should show that all relevant interfaces are up and functioning correctly, indicating successful resolution of the optical transport failure.

Command: display optical-attribute, Expected Output: Optical attributes within normal ranges, Description: Verifying the optical attributes again should confirm that the transmit and receive power levels are within the expected ranges, ensuring that the optical signal is being transmitted correctly.
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Roll-back Procedures

- Step: Revert SFP/XFP Replacement, Command: replace sfp slot with original, Description: If replacing the SFP/XFP does not resolve the issue or introduces new problems, reverting to the original module may be necessary. This involves physically removing the new SFP/XFP and reinstalling the original one.

- Step: Reconnect or Repair Fiber, Command: None (Physical Intervention Required), Description: If adjustments or repairs to the fiber do not resolve the issue, reconnecting or further repairing the fiber segment may be required. This step, like the initial repair, does not involve CLI commands and requires physical intervention.

- Step: Reset Rx/Tx Power Levels, Command: reset optical-power , Description: If adjusting the power levels does not resolve the issue, resetting them to their default settings may be necessary. The interface should be specified according to the device's documentation and the results of the diagnostic commands.